

Question	Answer	Mark	Mark scheme	Additional guidance
15 (a)	$(a - b)(a + b)$	B1	for $(a - b)(a + b)$ oe	Note that 1048575 and 1048577 earns 2 marks as an alternative approach
(b)	Relationship shown	M1	for factorising the expression, eg $(2^{20} - 1)(2^{20} + 1)$ or for $(2^{40} =) (2^{20})^2$	
		C1	for explanation, eg 2^{20} is even so $2^{20} - 1$ and $2^{20} + 1$ are odd (leading to conclusion)	